

Hypothesis Testing with E-values - by A. Ramdas and B. Wang

Reading Group 2025

Chapter 3 - E-values
under the alternatives
hypothesis

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0 - Introduction : Neyman-Pearson framework

Measurable space $(\Omega, \mathcal{F})$ $\mathcal{M}_1 = \{ \text{prob on } (\Omega, \mathcal{F}) \}$

$X_1, X_2, \dots$ generate according to $P^+ \in \mathcal{M}_1$

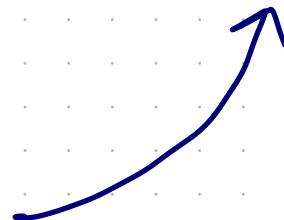
$$\mathcal{H}_0 : P^+ \in \mathcal{P} \subset \mathcal{M}_1 \text{ v.s. } \mathcal{H}_1 : P^+ \in \mathcal{Q} \subset \mathcal{M}_1$$

Test Reality \	H_0	H_1
H_0	✓	✗
H_1	✗	✓

Type I error



Type II error



Seen last week with François

Test \ Reality	H_0	H_1
H_0	✓	✗
H_1	✗	✓

$$P(\text{Type I error}) \leq \alpha$$

where α is the level of the test

Today

Type II error

Goal : Describe the desired properties of an ϵ -value under the alternative

Two notions of power

I) Power $\rightarrow$ classical notion in hypothesis testing

II) ϵ -power $\rightarrow$ specific to ϵ -value

I) Power

Powered tests and e-variable

Def 3.1

- The power of a (non-binary) test against an alternative Q is $E^Q[\phi]$

The type II error of ϕ equals
 $1 - E^Q[\phi]$

- When testing a null hypothesis $\mathcal{P}$ a test ϕ is powered at level $\alpha \in (0, 1)$ against $\mathcal{Q}$ if

$$E^{\mathbb{P}}[\phi] \leq \alpha \quad \text{for every } \mathbb{P} \in \mathcal{P}$$

$$\underline{\text{and}} \quad E^{\mathbb{Q}}[\phi] > \alpha \quad \text{for every } \mathbb{Q} \in \mathcal{Q}$$

It's uniformly powered if further

$$\inf_{\mathbb{Q} \in \mathcal{Q}} E^{\mathbb{Q}}[\phi] > \alpha$$

• An e-variable E is powered against
 $\mathcal{Q}$ if

$$\mathbb{E}^Q[E] > 1 \text{ for every } Q \in \mathcal{Q}$$

It's uniformly powered against $\mathcal{Q}$ if

$$\inf_{Q \in \mathcal{Q}} \mathbb{E}^Q[E] > 1$$

The existence of powered test or e-variable require that the alternative $\mathcal{Q}$ doesn't contain a "boundary point" of the null $\mathcal{P}$

$$H_0 : \mu \leq 0 \quad \text{v.s.} \quad H_1 : \mu > 0 \quad \text{ok}$$

$$H_0 : \mu < 0 \quad \text{v.s.} \quad H_1 : \mu \geq 0 \quad \text{not ok}$$

Connection between e-value and tests

Def 3.5 An e-variable E is called an all-or-nothing e-variable if

$$E = \frac{1}{c} \times 1_A$$

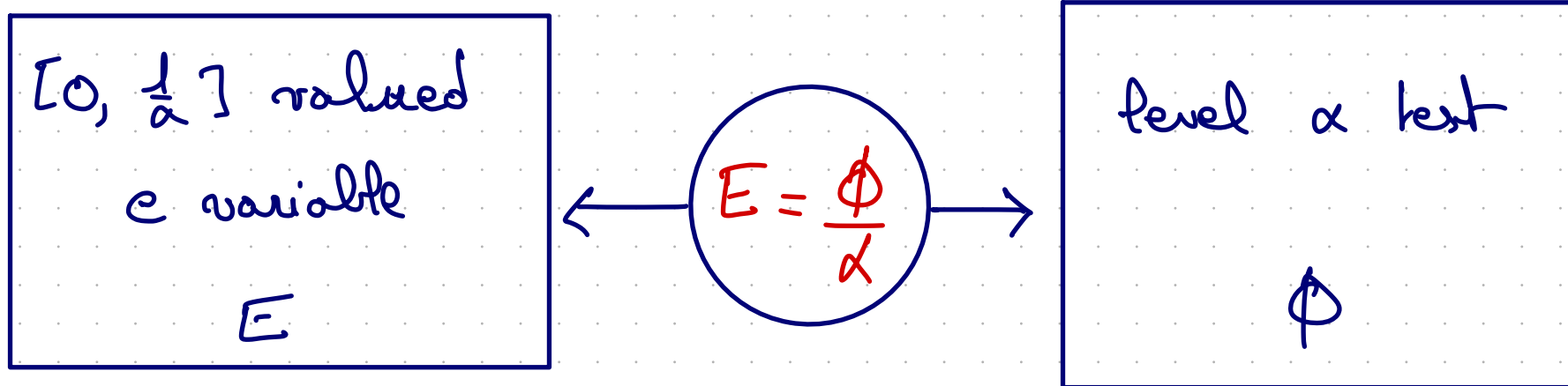
for some event $A \in \mathcal{F}$ and $c > 0$.

It's clear that A must satisfy

$$\mathbb{P}(A) \leq c \quad \forall \mathbb{P} \in \mathcal{P}$$

Fact 3.5

- For any $\alpha \in (0, 1)$, there is a one-to-one correspondence between



- If ϕ is binary then E is an all-or-nothing e-variable taking value $\{0, \frac{1}{\alpha}\}$
- Thus there exist a powered level α binary test iff there exists a powered e-variable taking value $\{0, \frac{1}{\alpha}\}$

- For any binary test ϕ that reject the null when a realized p value is smaller than or equal to α . ϕ can be expressed in terms of the all-or-nothing e -variable $\mathbb{1}_{\{P \leq \alpha\}} \sim \frac{1}{\alpha}$

ie the tests routinely use in science can be expressed with e -value

- A non binary test ϕ can be converted into a binary test ϕ' by taking

$$\phi' = \mathbb{1}_{\{u \leq \phi\}} \quad u \sim \mathcal{U}[0,1]$$

- An e variable taking value in $[0, 1/\alpha]$ is associated with the level α test $\Phi = \alpha E$ but also yields a level β test for $\beta \leq \alpha$.
So a single e -variable yields a family of test

Conclusion :

e -values suffice for testing

If a problem is testable (i.e. a powered test exists) then it is testable with e -value

A unified existence result

Th 3.9 For testing P against Q , the following are equivalent

- ▶ there exist a bounded powered e-variable
- ▶ there exist a powered - level α test for some (or every) $\alpha \in (0, 1)$
- ▶ there exist a powered - level α binary test for some (or every) $\alpha \in (0, 1)$

Condition for the existence of powered e-variable

Th 3.17 For testing $\mathcal{P}$ against $\mathcal{Q}$, with a common reference measure, the following are equivalent

► There exist a test ϕ st

$$\inf_{Q \in \mathcal{Q}} \mathbb{E}^Q[\phi] > \sup_{P \in \mathcal{P}} \mathbb{E}^P[\phi]$$

► There exist a uniformly powered bounded e-variable E for $\mathcal{P}$ against $\mathcal{Q}$

$$\sup_{\substack{P \in \text{Conv}(\mathcal{P}) \\ Q \in \text{Conv}(\mathcal{Q})}} d_{TV}(P, Q) > 0$$

II) E - power

The classical notion of power makes sense only for bounded c-variable through their link with ket :

We need a notion which work for
all c-variable

Def 3.14

- The e-power of an e-variable E against a simple alternative $\mathbb{Q}$ is

$$E^{\mathbb{Q}}[\log E]$$

assuming that the expectation is well defined

- An e-variable has positive e-power against $\mathbb{Q}$ if

$$E^{\mathbb{Q}}[\log E] > 0 \quad \forall \mathbb{Q} \in \mathcal{Q}$$

- It has uniformly positive e-power

against $\mathbb{Q}$ if

$$\inf_{\mathbb{Q} \in \mathcal{Q}} E^{\mathbb{Q}}[\log E] > 0$$

- There is some axiomatic justification for this notion of e-power $\rightarrow$ see section 3.8
- The e-power of an all-or-nothing e-variable is $-\infty$

Relating e-power with power



Prop 3.12 Let $E_1, E_2, \dots$ be a sequence of e-variables that are iid under $\mathbb{Q}$ and define

$$\Pi_t = \prod_{s=1}^t E_s \quad \forall t \in \mathbb{N}$$

If $E^{\mathbb{Q}}[\log E_1] > 0$ then $\forall \alpha > 0$

$$\mathbb{Q}(\Pi > \frac{1}{\alpha}) \xrightarrow[t \rightarrow \infty]{} 1$$

If $E^{\mathbb{Q}}[\log E_1] < 0$ then $\forall \alpha > 0$

$$\mathbb{Q}(\Pi > \frac{1}{\alpha}) \xrightarrow[t \rightarrow \infty]{} 0$$

Remember that π_k is an e-variable $\forall k$

Let's convert π_k into a level α test:

$$\Phi = \mathbb{1}\{\pi_k > \frac{1}{\alpha}\}$$

then a positive e-power leads to a test with asymptotic power 1 and a negative power leads to asymptotic power 0.

• By Jensen $E^Q[\log E] > 0 \Rightarrow E^Q[E] > 1$

so if E has positive e-power then E is a powered e-variable.

The reverse is not true but this two notions are related:

Th 3.14 For a simple alternative Q for any powered e-variable E the e-variable $1 - \lambda + \lambda E$ has positive e-power for some λ small enough

NB

- The e-power $E^Q[\log E]$ may not always be well-defined
- In some special case the e-power although well-defined, may be infinite, and does not have a useful interpretation

Log-optimality and likelihood ratio

Def 3.18 When comparing two e-variables E_1 and E_2 for $\mathcal{P}$ against simple $\mathcal{Q}$, E_1 is said to have greater (resp. equals) e-power than E_2 if

$$\mathbb{E}^{\mathcal{Q}} \left[\log \left(\frac{E_2}{E_1} \right) \right] \leq 0 \quad (\text{resp. } = 0)$$

Moreover, E is log-optimal if $\mathbb{E}^{\mathcal{Q}} \left[\log \left(\frac{E'}{E} \right) \right] \leq 0$ for any other E variable E'

- $E^Q \left[\log \left(\frac{E'}{E} \right) \right] \leq 0 \iff E^Q [\log E'] \leq E^Q [\log E]$

when E and E' have finite e -power
 (but not in general)

Existence of a log optimal e -variable.

Prop 3.19 For testing any $\mathcal{P}$, a log-optimal e -variable against Q always exist and is Q almost surely unique.

Th 3.20 (e-analogy of the Neyman-Pearson Lemma)

For testing simple $\mathbb{P}$ against simple $\mathbb{Q}$ the
($\mathbb{P}$ a.s.) unique log-optimal e-variable is the
likelihood ratio $\frac{d\mathbb{Q}}{d\mathbb{P}}$

The the optimal e-power equals

$$KL(\mathbb{Q}, \mathbb{P}) = E^{\mathbb{Q}} \left[\log \frac{d\mathbb{Q}}{d\mathbb{P}} \right]$$

which can be infinite

Conditional log-optimality

Usually $\mathcal{E}$ -variables are constructed based on available data $(X_1, \dots, X_n)$ and thus belong to a sub- σ -algebra $\mathcal{G} \subset \mathcal{F}$ generated by the data.

Def A log-optimal $\mathcal{E}$ -variable with respect to $\mathcal{G}$ is an $\mathcal{E}$ -variable E measurable with respect to $\mathcal{G}$ that satisfies $E^Q[\log(\frac{E'}{E})] \leq 0$ for all $\mathcal{E}$ -variable E' that are measurable with respect to $\mathcal{G}$.

We note $\forall P \in \mathcal{A}$ $P|_{\mathcal{G}}$ the restriction of P to $\mathcal{G}$ that is $P|_{\mathcal{G}}(A) = P(A) \quad \forall A \in \mathcal{G}$

Prop 3.22 Let $\mathcal{G}$ be a sub- σ -algebra of $\mathcal{F}$.

If $Q \ll P$, then the log-optimal ϵ variable for testing P against Q with respect to $\mathcal{G}$ exist and is given P almost uniquely by

$$E = \frac{dQ|_{\mathcal{G}}}{dP|_{\mathcal{G}}} = E^P \left[\frac{dQ}{dP} | \mathcal{G} \right]$$

A symmetric log-optimality

Before any $\mathcal{P}$ against simple $\mathcal{Q}$

Now simple $\mathbb{P}$ against composite $\mathcal{Q}$

$X^n = (X_1 \dots X_n)$ the data

A reasonable objective is the asymptotic log optimality i.e. define a sequence of e-variable

by $E_n = E(X^n)$

such as for any $\mathbb{Q} \in \mathcal{Q}$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E}^{\mathbb{Q}|_{\mathcal{F}(X^n)}} [\log E_n] = KL(\mathbb{Q}, \mathbb{P})$$

$KL(Q, P)$ = e power of an oracle that knows which alternative distrib is generating the data

If the limit holds, our e-variable is able to asymptotically achieve the same power than an oracle

Seems cool! How can we build asymptotic log optimal e-variable?

Mixture method

Pick a mixture distrib J over $\mathcal{Q}$ (with full support) and define

$$E_n = \int \prod_{i=1}^n \frac{q(X_i)}{p(X_i)} J(dq)$$

With some assumption on J $(E_n)_n$ is asymptotically log optimal

Plug in method

Let $Q_{i-1} \in \mathcal{Q}$ be a distri of the i -th observation, picked based on the first $i-1$ observations, and q_{i-1} its density. Define

$$E_n = \prod_{i=1}^n \frac{q_{i-1}(X_i)}{p(X_i)}$$

is asymptotically log optimal for a good choice of Q_{i-1} for instance

the posterior mean of $Q \in \mathcal{Q}$ after seeing X^{i-1} when starting with prior $\mathcal{Q}$.

III) Summary

Two notions of power

power

$$\mathbb{E}^Q[E]$$

= classical (test)
power for
bounded e-
variable

e-power

$$\mathbb{E}^Q[\log(E)]$$

- allow to compare any e-variable
- asymptotic interpretation in term of test